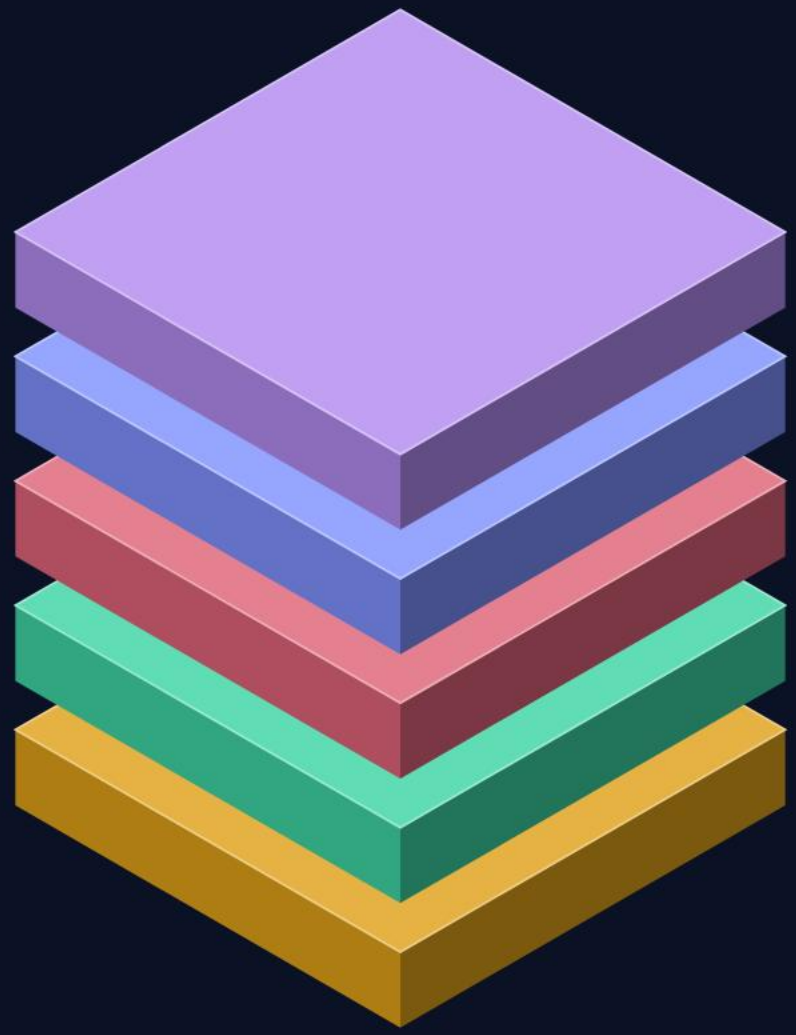
HACKATHON CHALLENGE 11

Fraud Hunter India

Stops the scam in about a millisecond, and explains why in English and Hindi.

Abhishek Kumar



THE PROBLEM

Asha is not hacked. She is talked into paying.

9:00 pm, an SMS

Your electricity will be cut tonight. Call this number now.

The call

A polite officer asks for a Rs 10 check and a support app. It is a screen-sharing tool.

The payment

While Asha watches, Rs 18,500 leaves her UPI app to a stranger. A mule account.

A schoolteacher in Patna. Rs 18,500 gone in seconds. The old system saw an amount that was large for her, but not impossible, and approved it.

WHY OLD SYSTEMS FAIL

Two ways to lose, and most systems pick one.

Approve the scam

Looks at one thing: is the amount unusual for this customer? A first-time payee plus a screen-sharing app does not show up.

Block the good customer

Turn the dial up and genuine payments get stopped. Customers leave, support queues grow, and trust falls.

Blocking a good customer is also a loss. A real system has to price both mistakes.

THE IDEA

Not “is this fraud?” but “what is the cheapest safe action?”

APPROVE

Let it through, quietly.

Rs 249 to a known merchant.

STEP-UP

Ask for more proof first.

A large payment to a new person, no other warning.

DECLINE

Stop it and say how to report.

Rs 18,500 to a first-time person with a screen-share running.

The bar moves with the amount: fraud on Rs 90,000 costs more than an extra check on Rs 900.

HOW IT WORKS

Five steps, about one millisecond.

1
Memory
44 live signals per customer, no peeking ahead

→

2
Model
Additive, so every reason is exact arithmetic

→

3
Guardrails
Rules that can only make it stricter

→

4
Cost
Expected cost of each action

→

5
Answer
Decision plus reasons in Hindi and English

Median 0.10 ms per decision on the simulator. One worker, because the customer memory lives in process.

THE REAL OUTPUT

Asha's payment, replayed.

```
Declined at 98.0% fraud risk. Rule SCREEN_SHARE_NEW_PAYEE fired.

Drivers: screen-sharing or remote-access app active; first payment to this payee;
payee is a person or account, not a verified merchant.

Expected cost if approved Rs 18,716 vs Rs 8 for the chosen action.
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The customer sees the reason in Hindi or English and is told to call 1930, the cyber-crime helpline.

BEYOND ONE PAYMENT

Two tools for what a single payment cannot show.

Scam Shield

Paste a suspicious SMS before paying. 26 rules plus a small learned word model, scored the same in the browser and on the server.

Catches things like “your electricity will be cut tonight”.

Mule graph

Links accounts by shared devices and payees to find the ones that only pass stolen money on.

Payee reputation that forgets, so a reformed account is not blamed forever.

THE CONSOLE

Eleven tabs you can click.

Live ops · Review queue · Backtest · Red team · Policy lab · Playground · Model card · Scam Shield · Mule graph · Live wire · Real data

Try it: press Launch attack on Live ops, paste an SMS into Scam Shield, then open Real data.

PROOF, WITH ITS LIMITS

Tested on real public data.

0.98

AUC on real card fraud

284,807 real payments, 492 fraud

64%

real scam texts flagged

Held-out half. 56% on wording never seen before.

0.04%

false alarms on real texts

Out of about 27,000 normal messages

0.10

ms median decision

On the simulator, not real traffic

Green is real public data. Amber is the simulator. The card data has only 75 fraud cases in its test set.

WHAT IS AND IS NOT REAL

Every tab says which one it is.

Real public data	The model on 284,807 real card payments; Scam Shield on about 57,000 real texts.
Real test mode	Signature-checked Razorpay test payments flowing through the engine.
Simulated	All payment data in the simulator tabs: eight Indian fraud types, twelve look-alike legitimate behaviours.
Not done	Real money, a real bank's customers, and evidence on real customer-level fraud.

BUILT TO RUN

It is a service, not just a demo.

Shadow mode

Scores real traffic and blocks nothing, so a bank can measure first.

Backtest

Replay your own CSV history and see cost, fraud caught and customers touched.

Safe by default

Per-IP rate limits, signed webhooks, API key on heavy jobs.

One image, two clouds

Docker on Render and on AWS ECS, with 154 automated tests.

THE ASK

Run it beside your system. Measure first. Trust later.

GitHub: github.com/antrikakashyap/fraud_hunter

Live console: fraud-hunter-api.onrender.com

Video: docs/demo.mp4 in the repo

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